

Lab 8: Implementing Cost Optimization Strategies

Duration: 30 minutes · **Region:** us-east-2 (Ohio)

Objectives

- Compare On-Demand and Savings Plan pricing for your web server
 - Create a budget alert
 - Rightsize an instance
 - Clean up everything so nothing is billed overnight
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Exercise 8.1: Price It (6 min)

1. Open <https://calculator.aws> → **Create estimate** → **Add service** → **Amazon EC2** → **Configure**.
2. Region **US East (Ohio)**, Linux, instance **t3.micro**, quantity **2**, utilization 100%.
3. **Payment option: On-Demand** → note the monthly cost.
4. Change to **Compute Savings Plans, 1 year, No upfront** → note the new monthly cost.

Same servers, ~30% cheaper for a one-year commitment. Commit for the steady baseline; stay On-Demand for spikes and experiments.

Exercise 8.2: Budget (6 min)

1. Search **Billing** → **Budgets** → **Create budget** → **Customize (advanced)** → **Cost budget** → **Next**.
2. Name `userN-budget`, Monthly, Fixed, amount **10 USD** → **Next**.
3. **Add alert threshold:** 80%, **Actual**, your email → **Next** → **Next** → **Create budget**.

Exercise 8.3: Rightsize (6 min)

`userN-web-2` is oversized for a page that serves one line of HTML.

1. **EC2** → **Instances** → select `userN-web-2` → **Instance state** → **Stop instance**. Wait for *Stopped*.
2. **Actions** → **Instance settings** → **Change instance type** → **t3.nano** → **Apply**.
3. **Instance state** → **Start instance**. Browse to its (new) public IP — same page, half the price.

Exercise 8.4: Clean Up (12 min)

Work top to bottom.

☐	Resource	Where	How
☐	userN-web-1 , userN-web-2	EC2 → Instances	Instance state → Terminate
☐	userN-web-ami	EC2 → AMIs	Actions → Deregister AMI → check <i>Delete associated snapshots</i>
☐	userN-key	EC2 → Key Pairs	Delete
☐	userN-db	RDS → Databases	Actions → Delete → uncheck <i>final snapshot</i> and <i>retain backups</i> → type delete me
☐	userN-db-subnets	RDS → Subnet groups	Delete (after the database is gone)
☐	userN-hello	Lambda	Actions → Delete
☐	userN-bucket-XXXX	S3	Empty first, then Delete
☐	userN-vpc	VPC → Your VPCs	Actions → Delete VPC (removes subnets, IGW, security groups). If refused, the RDS instance is still deleting — wait and retry.
☐	userN-budget	Billing → Budgets	Delete
☐	userN-dev , userN-developers , userN-ec2-role	IAM	Delete each

Final check: EC2 Instances shows only *terminated* rows with your name; RDS and S3 show nothing of yours.



Knowledge Check

1. Which option would you pick for a server that runs 24×7 for three years? For a two-hour experiment?
2. What would still cost money if you had *stopped* the instances instead of terminating them?